Chapter 11 — Hydrocarbons

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. Hydrocarbons contain only:

(a) H and O **(b)** C and N **(c)** C and O **(d)** C and H

2. Alkanes have general formula:

(a) C_nH_{2n} **(b)** C_nH_n **(c)** C_nH_{2n-2} **(d)** C_nH_{2n+2}

3. Cycloalkanes have formula:

(a) C_nH_{2n-2} **(b)** C_nH_n **(c)** C_nH_{2n} **(d)** C_nH_{2n+2}

4. Alkenes contain:

(a) C≡C only **(b)** Benzene only **(c)** C-C only **(d)** C=C

5. Alkynes contain:

(a) Aromatic ring **(b)** C≡C **(c)** C=C **(d)** C-C only

6. Benzene formula:

(a) C_6H10 **(b)** C_6H_{14} **(c)** C_6H12 **(d)** C_6H_6

7. IUPAC parent chain is:

(a) Longest carbon chain **(b)** Most branched only **(c)** Shortest chain **(d)** Ring always

8. Substituents get:

(a) Random numbers **(b)** No numbers **(c)** Highest locants **(d)** Lowest locants

9. Homolytic fission gives:

(a) Protons **(b)** Free radicals **(c)** Ions **(d)** Neutrons

10. Heterolytic fission gives:

(a) Ions **(b)** Radicals **(c)** Molecules only **(d)** Atoms only

11. Chlorination of methane needs:

(a) Darkness **(b)** UV light **(c)** Low temperature only **(d)** No initiator

12. Initiation step forms:

(a) HCl only **(b)** Cl radicals **(c)** CH3Cl **(d)** NaCl

13. Termination step:

(a) Radicals form **(b)** Chain grows **(c)** UV absorbed **(d)** Radicals combine

14. Alkanes are unreactive toward polar reagents due to:

(a) Non-polar strong bonds **(b)** Weak bonds **(c)** Ionic bonds **(d)** Pi bonds

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15. Complete combustion of alkane gives:

(a) CO_2 and H_2O **(b)** H_2 only **(c)** CO only **(d)** C only

16. Alkenes mainly undergo:

(a) Nucleophilic substitution **(b)** Free-radical substitution **(c)** No reaction **(d)** Electrophilic addition

17. Markovnikov rule: H adds to carbon with:

(a) More hydrogens **(b)** Fewer hydrogens **(c)** More alkyl groups **(d)** No hydrogens

18. Most stable carbocation:

(a) Methyl **(b)** Primary **(c)** Secondary **(d)** Tertiary

19. Alkyl groups show:

(a) No effect **(b)** +I effect **(c)** -I effect **(d)** Mesomeric only

20. Hydration of ethene gives:

(a) Ethanoic acid **(b)** Ethanol **(c)** Ethyne **(d)** Ethane

21. Ozonolysis helps locate:

(a) Halogen **(b)** Ring size **(c)** OH group **(d)** Double bond

22. Benzene prefers:

(a) No reaction **(b)** Electrophilic substitution **(c)** Free-radical addition **(d)** Electrophilic addition

23. Huckel rule pi electrons:

(a) 4n **(b)** 2n **(c)** 4n+2 **(d)** n+2

24. Benzene C-C bonds are:

(a) Triple **(b)** Alternating long/short **(c)** Ionic **(d)** Equal length

25. Chain isomerism differs in:

(a) Geometry **(b)** Position only **(c)** Carbon skeleton **(d)** Functional group

26. Position isomerism differs in:

(a) Position of group/bond **(b)** Molecular formula **(c)** Carbon count **(d)** Element type

27. Geometrical isomerism includes:

(a) Chain only **(b)** cis/trans **(c)** Metamerism only **(d)** Optical only

28. Conjugated dienes have:

(a) Only single bonds **(b)** Alternating double/single bonds **(c)** Only triple bonds **(d)** No pi bonds

29. Oxidation of ethanol can give:

(a) Benzene **(b)** Ethane **(c)** Ethanoic acid **(d)** Methane

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30. Reduction of carbonyl gives:

(a) Acid **(b)** Alcohol **(c)** Halide **(d)** Alkene

31. Hydrocarbons contain:

(a) C and N only **(b)** C and O only **(c)** C, H and O **(d)** C and H only

32. Aliphatic hydrocarbons are:

(a) Ionic only **(b)** Open-chain or non-aromatic cyclic **(c)** Metallic only **(d)** Benzene only

33. Aromatic hydrocarbons are based on:

(a) Metal bond **(b)** Benzene ring **(c)** Ionic lattice **(d)** Open chain only

34. Alkenes have general formula:

(a) C_nH_{2n-2} **(b)** C_nH_{2n+2} **(c)** C_nH_n **(d)** C_nH_{2n}

35. Benzene molecular formula is:

(a) C_6H12 **(b)** C_6H_6 **(c)** C_6H_{14} **(d)** C_2H_2

36. Cycloalkanes have formula:

(a) C_nH_n **(b)** C_nH_{2n-2} **(c)** C_nH_{2n+2} **(d)** C_nH_{2n}

37. IUPAC parent name comes from:

(a) Longest carbon chain **(b)** Most branched chain **(c)** Random chain **(d)** Shortest chain

38. Substituents are numbered to give:

(a) No locants **(b)** Random locants **(c)** Highest locants **(d)** Lowest locants

39. Homolytic fission gives:

(a) Free radicals **(b)** Atoms only **(c)** Ions **(d)** Molecules only

40. Heterolytic fission gives:

(a) No species **(b)** Ions **(c)** Free radicals **(d)** Atoms only

41. Initiation step produces:

(a) Products only **(b)** Radicals **(c)** Ions only **(d)** No change

42. Propagation step:

(a) Forms ions only **(b)** Stops chain **(c)** No radical **(d)** Consumes and regenerates radical

43. Termination step:

(a) Combines radicals **(b)** Creates radicals **(c)** Only removes H_2 **(d)** Only adds H_2

44. Markovnikov addition puts H on carbon with:

(a) No hydrogens **(b)** More hydrogens **(c)** More halogens **(d)** Fewer hydrogens

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45. Most stable carbocation is:

(a) Tertiary **(b)** Secondary **(c)** Methyl **(d)** Primary

46. Alkyl groups show:

(a) -I effect **(b)** No effect **(c)** +I effect **(d)** -R only

47. Ozonolysis cleaves:

(a) O-H bond **(b)** C-C single bond only **(c)** C-H bond only **(d)** C=C bond

48. Conjugated dienes have:

(a) Only single bonds **(b)** Alternate double and single bonds **(c)** Only triple bonds **(d)** Isolated double bonds

49. Structural isomers have same:

(a) Structural formula **(b)** Density always **(c)** Molecular formula **(d)** Boiling point always

50. Benzene prefers:

(a) Electrophilic substitution **(b)** No reaction **(c)** Free radical addition **(d)** Electrophilic addition

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Answer Key with Brief Reasons

1. Answer: (d) C and H — Definition.

2. Answer: (d) C_nH_{2n+2} — Saturated acyclic.

3. Answer: (c) C_nH_{2n} — One ring.

4. Answer: (d) C=C — Double bond.

5. Answer: (b) C≡C — Triple bond.

6. Answer: (d) C_6H_6 — Aromatic hydrocarbon.

7. Answer: (a) Longest carbon chain — Nomenclature rule.

8. Answer: (d) Lowest locants — Nomenclature rule.

9. Answer: (b) Free radicals — Equal electron split.

10. Answer: (a) Ions — Unequal electron split.

11. Answer: (b) UV light — Initiation step.

12. Answer: (b) Cl radicals — Cl_2 -> 2Cl.

13. Answer: (d) Radicals combine — Ends chain.

14. Answer: (a) Non-polar strong bonds — C-C/C-H non-polar.

15. Answer: (a) CO_2 and H_2O — Full oxidation.

16. Answer: (d) Electrophilic addition — Pi bond electron-rich.

17. Answer: (a) More hydrogens — More stable carbocation.

18. Answer: (d) Tertiary — +I effect.

19. Answer: (b) +I effect — Electron releasing.

20. Answer: (b) Ethanol — Acid-catalysed addition.

21. Answer: (d) Double bond — Cleavage products.

22. Answer: (b) Electrophilic substitution — Aromatic stability.

23. Answer: (c) 4n+2 — Aromaticity.

24. Answer: (d) Equal length — Delocalization.

25. Answer: (c) Carbon skeleton — Structural isomerism.

26. Answer: (a) Position of group/bond — Same skeleton.

27. Answer: (b) cis/trans — Restricted rotation.

28. Answer: (b) Alternating double/single bonds — Delocalization.

29. Answer: (c) Ethanoic acid — Organic oxidation.

30. Answer: (b) Alcohol — Organic reduction.

31. Answer: (d) C and H only — Definition.

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32. Answer: (b) Open-chain or non-aromatic cyclic — Classification.

33. Answer: (b) Benzene ring — Stable ring system.

34. Answer: (d) C_nH_{2n} — One double bond.

35. Answer: (b) C_6H_6 — Aromatic hydrocarbon.

36. Answer: (d) C_nH_{2n} — Saturated ring.

37. Answer: (a) Longest carbon chain — Nomenclature rule.

38. Answer: (d) Lowest locants — Systematic naming.

39. Answer: (a) Free radicals — Equal electron split.

40. Answer: (b) Ions — Unequal electron split.

41. Answer: (b) Radicals — UV breaks Cl_2.

42. Answer: (d) Consumes and regenerates radical — Chain continues.

43. Answer: (a) Combines radicals — Chain ends.

44. Answer: (b) More hydrogens — More stable carbocation.

45. Answer: (a) Tertiary — +I effect.

46. Answer: (c) +I effect — Electron donating.

47. Answer: (d) C=C bond — Carbonyl fragments.

48. Answer: (b) Alternate double and single bonds — Delocalization.

49. Answer: (c) Molecular formula — Different connectivity.

50. Answer: (a) Electrophilic substitution — Preserves aromaticity.

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